
Golden-Section Search

Derivative-free optimization by interval reduction

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Four questions

- 1 Why does the constant $0.618\dots$ appear?
- 2 How can one function value be reused at every step?
- 3 What accuracy is guaranteed after a fixed budget?
- 4 When should we *not* trust the answer?

The working viewpoint

The method does not chase a derivative. It shrinks a bracket that is known to contain one optimum.

SECTION 1

Problem Setting

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A derivative-free question

$$x^* = \arg \max_{x \in [x_L, x_U]} f(x)$$

Only request: values of $f(x)$.

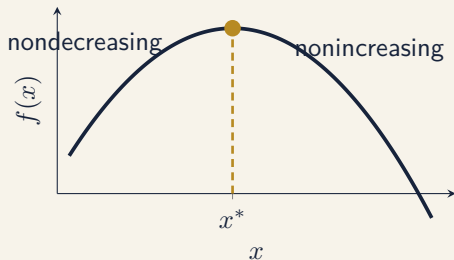
A reliable derivative may be unavailable, expensive, or meaningless.

Typical situations

- ▶ laboratory measurements;
- ▶ simulation runs;
- ▶ noisy or tabulated output.

For minimization, replace $\arg \max$ by $\arg \min$; the geometry is unchanged.

The unimodality contract



A maximum at x^* divides the bracket:

- ▶ f is nondecreasing up to x^* ;
- ▶ f is nonincreasing after x^* .

What is not required

No derivative, quadratic model, or smooth graph is required.

Source: Course notes, p. 1; maximization convention retained.

The bracket is a claim

- 1 **Valid domain:** every trial point is safe to evaluate.
- 2 **Contains the target:** the desired optimum is inside.
- 3 **One relevant mode:** one peak or trough governs the interval.

Why this matters

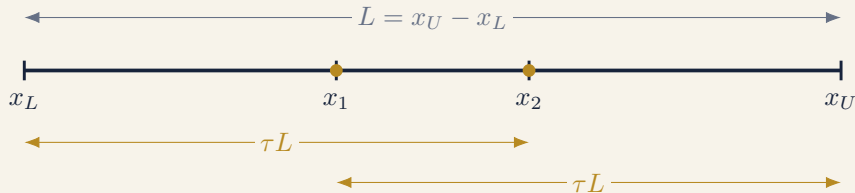
The method certifies a smaller *interval*; it cannot repair a bracket that excluded the optimum or mixed several extrema.

SECTION 2

The Search Mechanism

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Two probes, one bracket



$$x_1 = x_U - \tau L$$

$$x_2 = x_L + \tau L$$

$$\tau = \frac{\sqrt{5} - 1}{2} \approx 0.618034, \quad x_L < x_1 < x_2 < x_U.$$

Source: Course notes, pp. 1-2. The derivation of τ follows in the analysis section.

Which side survives?

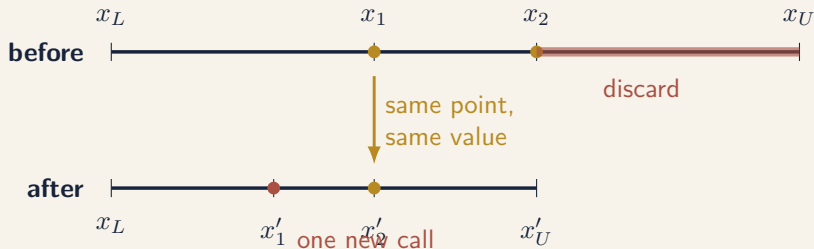
Comparison	Maximization keeps	Minimization keeps
$f(x_1) > f(x_2)$	$[x_L, x_2]$	$[x_1, x_U]$
$f(x_1) < f(x_2)$	$[x_1, x_U]$	$[x_L, x_2]$
$f(x_1) = f(x_2)$	$[x_1, x_2]$	$[x_1, x_2]$

Read one row, not the whole table

For a maximum, the larger value points toward the survivor. For a minimum, the smaller value does.

Source: The strict-inequality cases follow the update rule in the course notes, pp. 2-3.

Reuse one evaluation



For the illustrated maximization case, $f(x_1) > f(x_2)$. Therefore

$$x'_U \leftarrow x_2, \quad x'_2 \leftarrow x_1, \quad f(x'_2) \leftarrow f(x_1).$$

Only x'_1 is new, so only **one new function call** is needed.

Source: Course notes, p. 2.

Algorithm: initialize once

- 1 Set $\tau = (\sqrt{5} - 1)/2$ and $L = x_U - x_L$.
- 2 Form $x_1 = x_U - \tau L$ and $x_2 = x_L + \tau L$.
- 3 Evaluate $f_1 = f(x_1)$ and $f_2 = f(x_2)$.

State carried into the loop

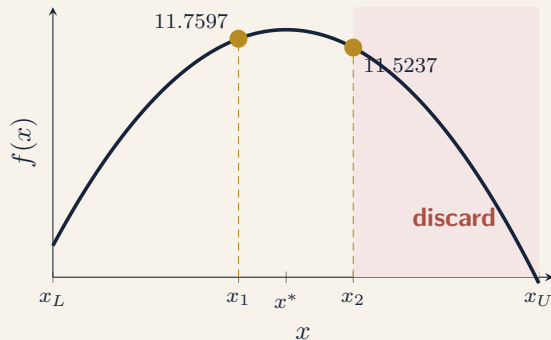
Keep the ordered endpoints, two probes, and their two function values. Losing a saved value also loses the main efficiency advantage.

Algorithm: repeat

- 1 Compare f_1 and f_2 .
- 2 Discard the side ruled out by unimodality.
- 3 Rename the surviving probe and reuse its value.
- 4 Create one new probe and evaluate f there.
- 5 Stop when the bracket meets the accuracy rule.

For minimization, reverse only the two strict comparison decisions; the probe geometry and reuse are unchanged.

Worked trace: first cut



Maximize

$$f(x) = 12 - (x - 2.4)^2$$

on $[0, 5]$.

$$x_1 = 1.90983, \quad x_2 = 3.09017.$$

Since

$$f(x_1) > f(x_2),$$

retain **$[0, 3.09017]$** .

Worked trace: cuts 0-2

k	Bracket before	x_1	x_2	Retained bracket
0	$[0, 5]$	1.90983	3.09017	$[0, 3.09017]$
1	$[0, 3.09017]$	1.18034	1.90983	$[1.18034, 3.09017]$
2	$[1.18034, 3.09017]$	1.90983	2.36068	$[1.90983, 3.09017]$

Watch the gold entry

It moves to the other probe column because the physical point and its function value are reused.

Worked trace: cuts 3-5

k	Bracket before	x_1	x_2	Retained bracket
3	[1.90983, 3.09017]	2.36068	2.63932	[1.90983, 2.63932]
4	[1.90983, 2.63932]	2.18847	2.36068	[2.18847, 2.63932]
5	[2.18847, 2.63932]	2.36068	2.46711	[2.18847, 2.46711]

Result after six comparisons

The true maximizer $x^* = 2.4$ remains inside $[2.18847, 2.46711]$, now only 0.27864 wide.

SECTION 3

Convergence and Cost

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SECTION 4

Reliable Computation

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SECTION 5

Mathematical Problems

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SECTION 6

Application Problems

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SECTION 7

Summary & References

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